Optuna: complete inverse-design procedure

Offline: prepare received HFSS data
2,000 spectra → 1,926 unique geometries
Group duplicates, normalize geometry

Retrain selected case topology from scratch
1,541 training / 385 validation / 0 test
Select final checkpoint by validation only

Load selected weights

User request
Wavelength band $[\lambda_{\min}, \lambda_{\max}]$ (nm)
Constant target absorption $A^\star$ (% → fraction)

Initialize all families F01–F12
8 training-only starts per family
Training bounds, local
trust radius: 10% of span

Evaluate frozen case surrogate
Family f + candidate geometry $\mathbf{p}$
Predict 401-point T/R/A
spectra (detail at right)

Evaluate requested absorption band
Interpolate exact band endpoints
$e_\infty = \max_j |\widehat{A}_j - A^\star|$
$e_{\text{mean}} = \frac{\int |\widehat{A}(\lambda) - A^\star| \, d\lambda}{\lambda_{\max} - \lambda_{\min}}$

Repeat

Update geometry only: 500 Adam steps
Gradient of $\text{mean}(e_\infty) + 0.001\,\text{mean}(e_{\text{mean}})$
Project to bounds, mask
and positive ring holes

After final evaluation

Rank candidates: e_∞, then e_{mean}
Best geometry in native nm: `geometry.json`
Predicted spectra and errors, pending HFSS

Run separate HFSS verification
Copy received template, apply selected air geometry
Preserve materials, boundaries and Floquet ports
Solve original 401-point sweep

Evaluate actual HFSS T/R/A
Compare actual absorption with request
Endpoint interpolation,
worst and mean errors

Expanded forward evaluation inside the optimization loop

Optuna: surrogate detail

Family-conditioned forward spectrum model (PyTorch)

Family index $f \in \{0, \ldots, 11\}$ (F01–F12)
Geometry $\mathbf{p} \in \mathbb{R}^7$, normalized by fixed scales

Embedding lookup $e(f) \in \mathbb{R}^{16}$
Family mask lookup $m(f) \in \{0,1\}^7$

Concatenate $[e(f),\ \mathbf{p} \odot m(f),\ m(f)]$
$16 + 7 + 7 = 30$ features

Linear $30 \to 256$ (bias) → SiLU

Block 1: Linear $256 \to 256$ (bias)
LayerNorm (256) → SiLU
Dropout $p = 0$ (identity)

Block 2: Linear $256 \to 256$ (bias)
LayerNorm (256) → SiLU
Dropout $p = 0$ (identity)

Block 3: Linear $256 \to 256$ (bias)
LayerNorm (256) → SiLU
Dropout $p = 0$ (identity)

Linear $256 \to 1203$ (bias)
Reshape: $1203 \to 401 \times 3$ logits

Softmax over T/R/A at each wavelength
$\widehat{\mathbf{Y}} \in [0,1]^{401 \times 3}$, $T_j + R_j + A_j = 1$

Batch dimension omitted. Plain blocks
are sequential, with no skip additions.